

5. Physical Organization

The computer memory is organised as main memory and Secondary memory.
main memory for fast access
Secondary memory for large storage.

Address binding

① Address binding is the process of mapping the program's logical or virtual addresses to corresponding physical or main memory addresses.

② Address binding is part of computer memory management and is performed by O.S on behalf of the applications that need access to memory.

③ Types of Address Binding

i) Compiled Time:- when the memory address is known at the compiled time then a code is directly converted into absolute code by the compiler.

ii) Load Time:- when the memory address is not known at the compiled time then binding is delayed till the load time, then compiler generates a relocatable code.

m) Execution Time:-

If a process can be moved from one segment of code to another during its execution then the address binding must be delayed till runtime, this requires a special hardware known as Memory Management unit.

Base Machine

- ① Base Machine is a logical hardware which is used to execute the program in the processor without using the operating system.
- ② Base machine only accepts instructions in m/c language.
- ③ Due to m/c lang. dependencies it omitted after arrival of O.S.

Resident Monitor.

- ① Resident Monitor is a code that runs on Base m/c.
- ② It works like job sequencer to sequence the jobs and sends them to the processor.
- ③ When the program execution occurs there is no gap between the program execution.

Fixed Size Memory partitioning

① It is a memory management technique ^{used in} ~~and~~ Computer System.

② Partition main memory into a set of non-overlapping memory regions called partitions.

③ Fixed partitions can be of equal or unequal sizes.

④ Each partition can hold exactly one process.

⑤ Advantages.

a) Simple Implementation b) Fast allocation

c) predictable performance.

⑥ Disadvantages.

a) Internal Fragmentation - If a partition is 100 MB and a process needs only 70 MB, the rest 30 MB is wasted, this is called Internal Fragmentation.

b) Limited flexibility

Process that need more memory than the size of partition leads to blockade due to mallocation.

c) Fixed no. of process

If all partitions are full, no more process can be loaded into memory.

Dynamic Partitioning

- ① It is memory management technique.
- ② memory is divided into partitions based on the needs of the processes rather than fixed sizes.
- ③ Advantages.
 - a) No Internal fragmentation :- means No wasted spaces.
 - b) Flexibility
 - c) Efficient memory utilization
- ④ Disadvantages.
 - a) Complex Implementation. → Requires algo to keep track of free and used memory.
 - b) Slower allocation. → due to searching of suitable block of memory that fits the process requirement.

Memory allocation

Memory allocation is the process by which a computer operating system assigns memory spaces to the various program and process that need to run.

Types

- ① Static allocation.

The space is fixed and can't be changed.

② Dynamic Memory Allocation.

Dynamic memory allocation is when a program can request memory while it is running rather than having memory before the program runs.

This allows programs to be flexible

Dynamic Memory Allocation Technique

① First fit.

① approach is to allocate the first free portion large enough to accommodate the process

② It finishes after finding the first suitable free portion.

working

① The allocator scans the memory block sequentially from the beginning

② It finds the first free block that is equal or large enough to accommodate the process

③ Memory is allocated to the requesting process.

② Best fit

① approach is to allocate the smallest free portion large enough to accommodate the process

② It finishes after finding the Best suitable free portion, aiming to minimize space waste.

working:

① The allocator scans all memory blocks to find the smallest block that meets the requirement.

⑤ The Smallest Suitable block is allocated to the requesting process.

⑤ Worst fit

① Approach is to locate largest available free portion ~~memory~~ to fulfill the request.

② It is just opposite of Best fit.

Working

1) The allocator scans all memory blocks to find the largest block.

2) The largest block is allocated to the requesting process.

Internal Fragmentation.

① Situation in which more memory is allocated than needed to the processes by the O.S is called Internal Fragmentation.

② Results in excess memory usage.

External Fragmentation

① The non-contiguous blocks creates free spaces in the memory resulting in unused storage that are outside the allocated regions.

② Hence the free spaces can't be used for larger memory task.

③ The Situation in which these free spaces ending up getting isolated and can't be totally eliminated from the memory space is called External Fragmentation.

④ Solution :- Compaction which shuffles contents of the memory to place all free memory together.

Q Given memory portions 100, 500, 200, 300, 600
Process allocation 212, 417, 112, 426

Process	Memory
P ₁	212
P ₂	417
P ₃	112
P ₄	426

First fit

Memory portion	Proc. NO.	Memory Req.	Status	Internal Fragmentation
100	-	-	Free	-
500	P ₁	212	Busy	288
200	P ₃	-	Free	88
300	-	112	Busy	-
600	P ₂	417	Busy	183

Total available: 1700

Total used: 141

Int. frag. : 559

Best fit

Mem. Partition	Proc. No.	Mem. Req.	Status	Int. Req.
100	-	-	F	-
500	P2	417	B	88
200	P3	192	B	88
300	P1	212	B	88
600	P4	426	B	174

Total avail = 1700

Total used = 1167

Int. Req = 433

Worst fit

Mem. Part.	Proc. No.	Mem. Req.	Status	Int. Req.
1000	-	-	F	-
500	P2	417	B	88
200	-	-	F	-
300	P3	112	B	188
600	P1	212	B	388

Total avail : 1700

Total used : 741

Int. Req : 659

Internal vs External.

Definition	wasted space within allocated memory blocks	wasted spaces outside of allocated memory blocks in free non-contiguous ^{memory} gaps.
Location.	occurs within the allocated memory of a process	occurs in the free memory spaces which are non-contiguous between allocated blocks
Cause	allocation in fixed size memory block	allocation and deallocation results non-contiguous memory gaps.
Solution.	Not fixed size memory block	compaction Memory compaction to create larger contiguous blocks of memory.
Memory		

Paging

- ① Paging is a memory management scheme that eliminates the need of contiguous allocation of physical address memory.
- ② Paging avoid external fragmentation and the need for compaction as it permits the physical address space of a process to be non-contiguous.
- ③ Paging divides the physical and process memory into fixed size blocks called pages (for process) and frames (for physical memory).

(1) Working

a) Divide memory

Both the process memory and physical memory are divided into equal sized pages and frames respectively.

b) Mapping pages to frames

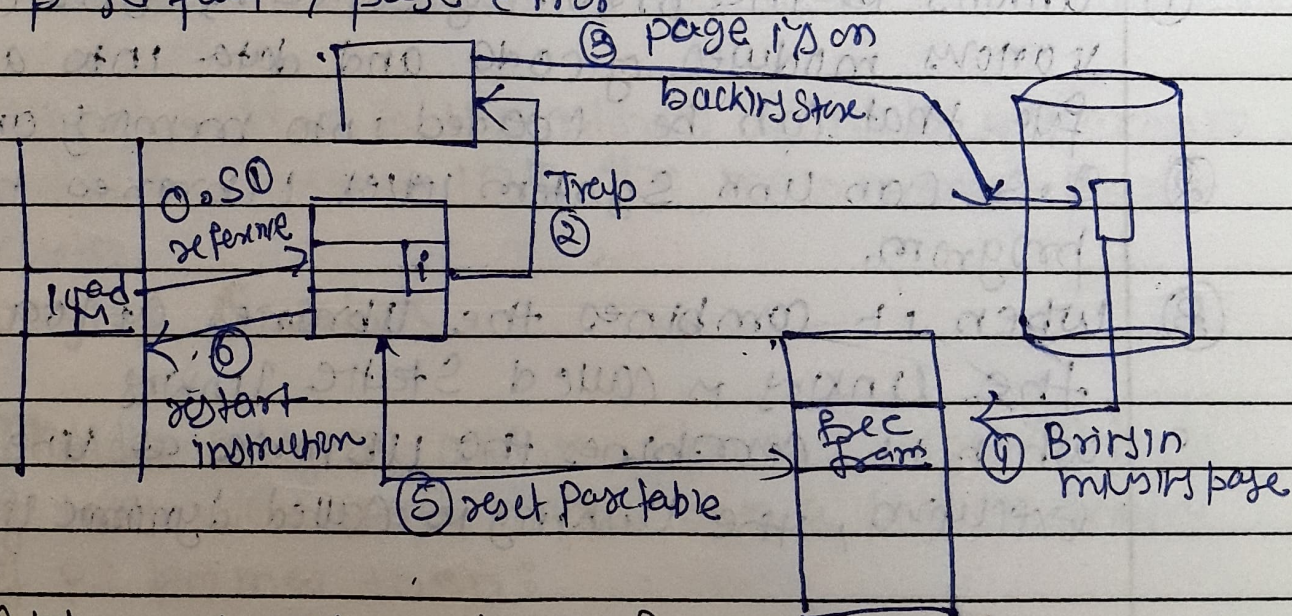
Each page of a process can be placed in any available frame of physical address.

c) Page table

A page table keeps tracks of ~~the~~ where each page of the process is stored in physical memory.

Page fault or page error.

- ① An interrupt that occurs when a program requests data that is not currently in real memory.
- ② The interrupt triggers the O.S to fetch the data from a virtual memory and load it into RAM.
- ③ A ~~the~~ ~~invalid~~ page fault or page error occurs when the O.S cannot find the data in virtual memory. Hence, we can say that when a page referenced by the program is not found in the main memory, it is called page fault / page error.



Steps of handling page fault →

Imp. Terminologies.

Dynamic Loading

- ① Refers to mapping an executable or library into process's memory after it has stored.
- ② In this a routine program is not loaded until it is called by the program.
- ③ makes better memory space utilization, Hence it is very useful when large amount of code are needed to handle in frequently occurring cases.

Dynamic Linking

- ① Linking is the process of collecting and combining various modules of code and data into a executable file that can be loaded into memory and executed.
- ② O.S. can link system level libraries to a program.
- ③ When it combines the libraries at loadtime, the linking is called Static Linking.
When it combines the libraries at time of execution, the linking is called dynamic linking.

Overlays

- ① It is used to keep instructions and data in memory that are needed at any given time.
- ② When other instructions are needed, they are loaded into space that was occupied previously by instructions that are no longer needed.

Swapping

- ① This makes it possible for the total physical address space of all processes to exceed the real physical memory of the system.
Hence it will increase the degree of multiprogramming.

Logical Address Space

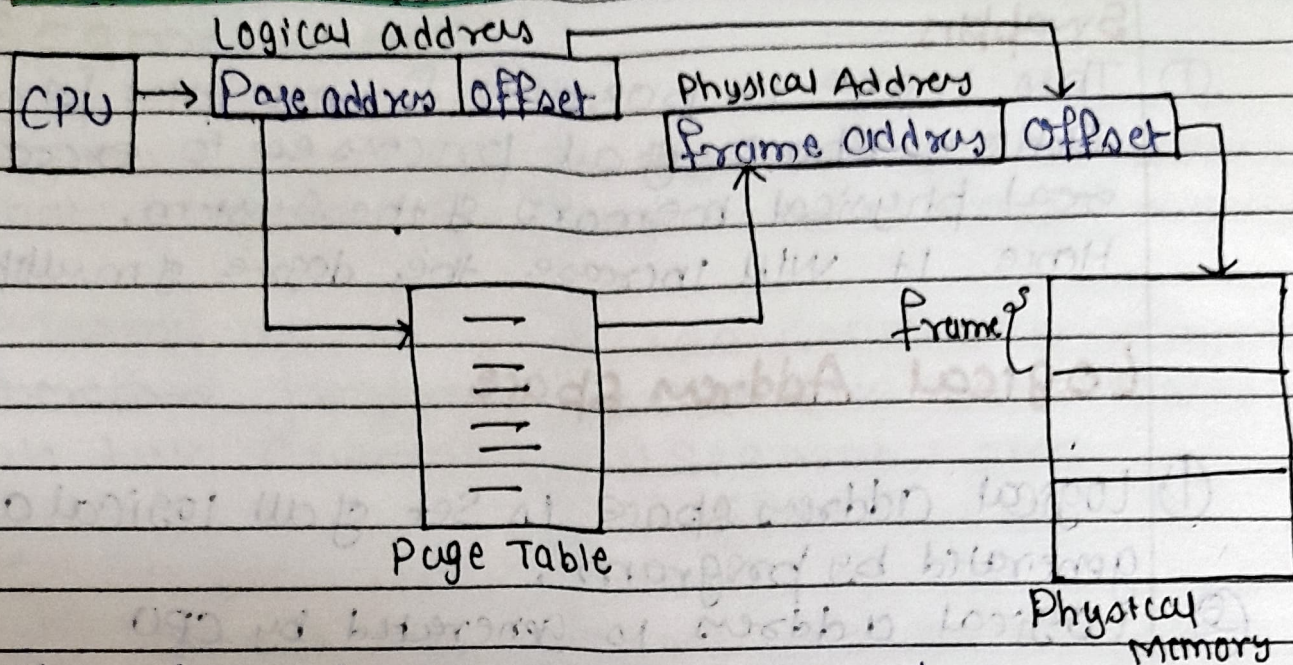
- ① Logical address space is set of all logical addresses generated by program.
- ② Logical address is generated by CPU.

Physical Address Space

- ① Physical Address space is a set of all physical addresses corresponding to these logical addresses.
- ② Physical address is the address of main memory and it is loaded into the main memory address register.

Paging Question

- Q Consider a logical address space of eight (8) pages of 1024 words each mapped onto a physical memory of 32 frames then:
 - a) How many bits are in logical address?
 - b) How many bits are in physical address?

Solⁿ

The Logical add. is split into 2 parts.
 (1) page add. and (2) page offset.

$$\text{Page add.} = 8 = 2^3 = 3 \text{ address bits}$$

$$\text{Page offset} = 1024 = 2^{10} = 10 \text{ address bits.}$$

$$\text{Total} = 13 \text{ address bits.}$$

The physical address is split into two parts
 (1) Frame address and (2) offset.

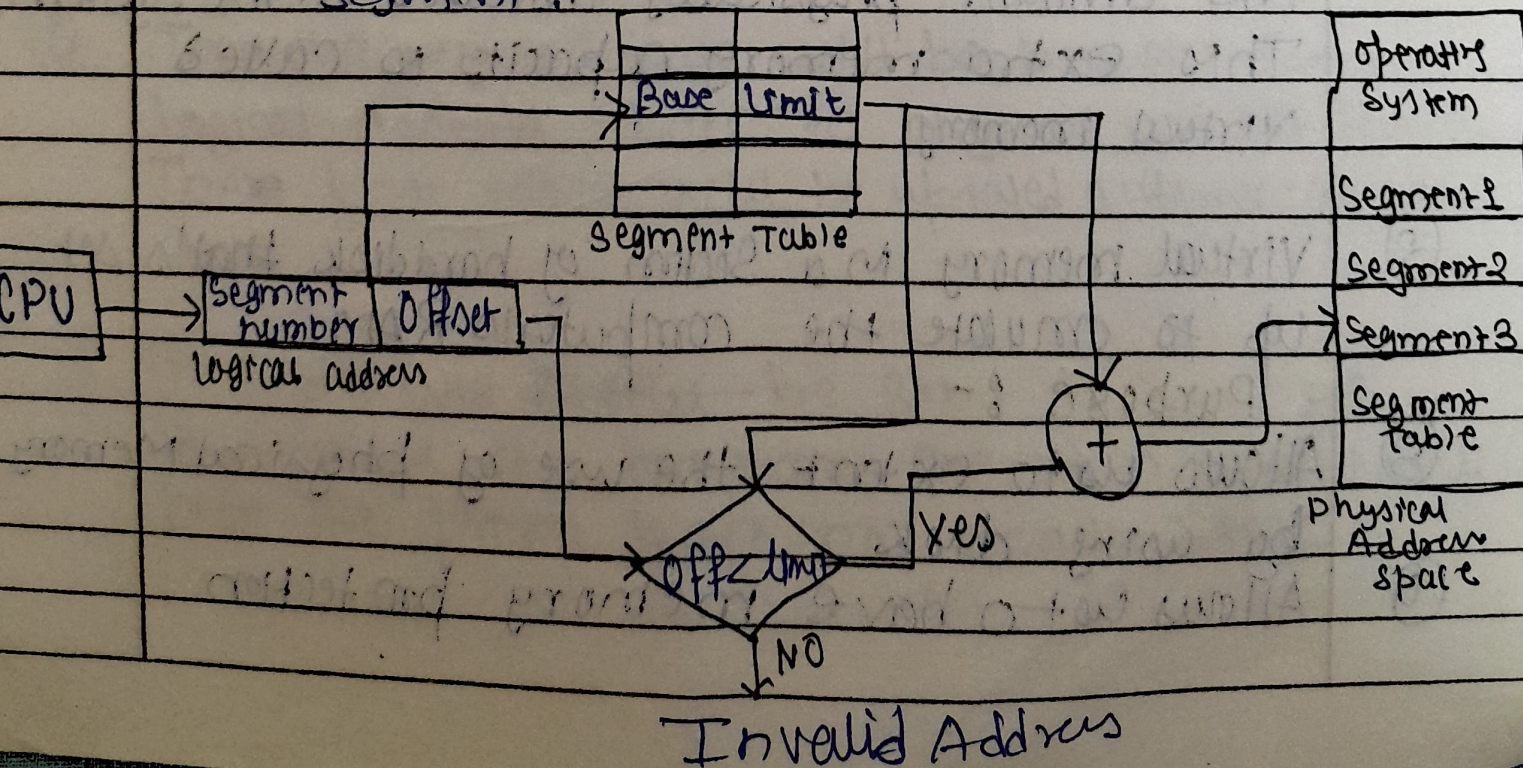
$$\text{Frame add.} = 32 = 2^5 = 5 \text{ address bits}$$

$$\text{Frame offset} = 2^{10} = 10 \text{ address bits.}$$

$$\text{Total} = 15 \text{ address bits.}$$

Segmentation.

- ① Segmentation is a memory-management scheme that supports the programmer view of memory.
- ② A logical address space is a collection of segments. Therefore, programmer specifies each address by two quantities, a) Segment name
b) offset.
- ③ For simple implementation,
 - a) Segments are numbered and are referred to as segment numbers.
 - b) Segment no. indexes a process segment table.
 - c) Each entry in segment table has
 - ① Segment base
 - ② Segment limit
 - d) Segment base contains physical address, where segment resides in memory.
 - e) Segment limit specifies the length of the segment.



Segmentation

Paging

- | | |
|---|--|
| ① Program is divided into Variable Size Segments. | ① Program is divided into Fixed Size pages |
| ② User is responsible for dividing program into segments. | ② O.S is responsible for dividing pages |
| ③ It is slower | ③ It is faster |
| ④ It is visible to user | ④ It is not visible to user |
| ⑤ It eliminates Internal Fragmentation. | ⑤ It eliminates external Fragmentation. |

Virtual Memory

- ① A computer can address more memory than the amount physically installed on the system. This extra memory capacity is called virtual memory.
- ② Virtual memory is a section of harddisk that's set up to emulate the computer's RAM.
Purpose :-
- ③ Allows us to extend the use of physical memory by using disk.
- ④ Allows us to have memory protection.

- ⑤ Virtual Memory is a memory management capability of an OS that uses hardware and software to allow a computer to compensate for physical memory shortage by temporarily transferring data from RAM to Disk Storage.

Demand Paging

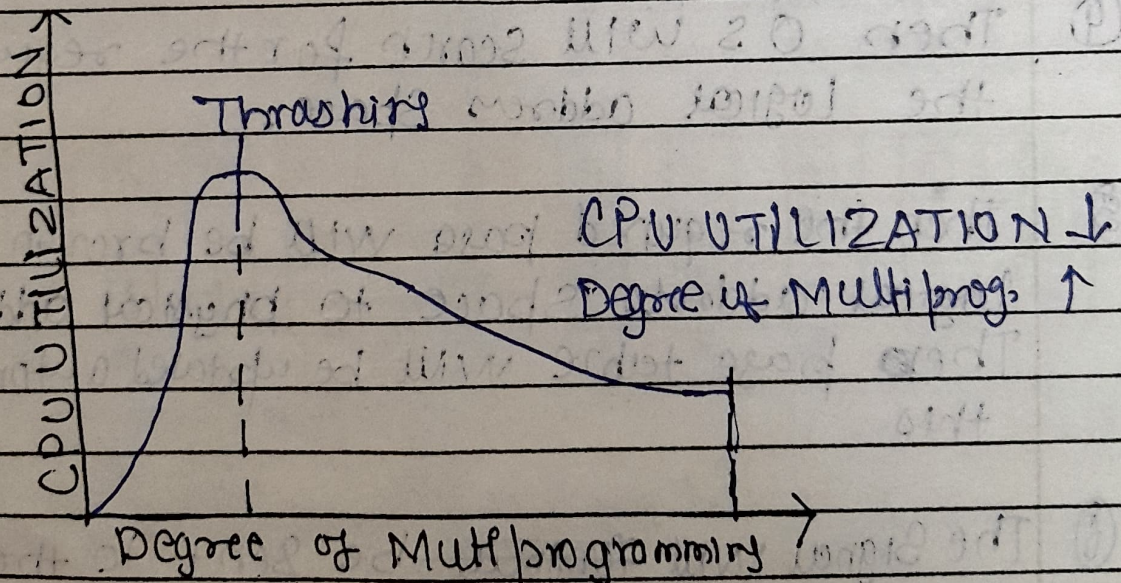
- ① The process of loading the page into memory on demand whenever page fault occurs is known as demand paging.

Working

- ② If CPU tries to refer a page that is currently not available in the main memory, it generates an interrupt indicating memory access fault.
- ③ The OS puts interrupted process in a blocking state.
- ④ Then OS will search for the required page in the logical address space.
- ⑤ Then the required page will be brought from logical address space to physical address space. The page table will be updated automatically upon this.
- ⑥ The signal will ~~transmit~~ be sent to the CPU to continue the program execution and it will place the process back into ready state.

Thrashing

- ① Thrashing occurs when a system spends more time in processing page fault rather than executing instructions.
- ② Thrashing is a discrete phenomenon occurs due to limited resources or exhaustion of memory.
- ③ Thrashing is busy swapping pages in and out i.e., doing more paging than executing.
- ④ The effect of Thrashing are:-
 - ① CPU becomes idle
 - ② CPU utilization decreases
 - ③ Degree of Multiprogramming increases



Page Replacement Algorithm

① First in First out [Fifo]

① Selects the pages for replacement that has been in memory for longest time

Reference String

6, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 1, 2, 0

Assume three frames and frames are initially empty

Symbol

Imp. Terminologies before soln

Hit (Pg)

① Page hit :- Whatever the page that is demanding in the reference string, if the page is present in the main memory, that is called page hit

* ② page fault :- Whatever the page that is demanded and it is not present is called page fault

Frame 1

" 2

" 3

6	6	6	2	2	2	2	9	4	4	0	0
	0	0	0	0	3	3	3	2	2	2	2
		1	1	1	1	0	0	0	3	3	3

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Hit (0)

Hit (3)

0	0	0
1	1	1
3	2	2

*

*

*

Hit (0)

Total page hit = 3

Total page fault = 12

Hit ratio $\Rightarrow \frac{\text{no. of hits}}{\text{Total no. of references}} = \frac{3}{15} \times 100$

Page fault Ratio = $\frac{12}{15} \times 100 \Rightarrow 80\%$

= 20%

② Least Recently Used (LRU)

① Selects the pages for replacement that has been in memory and least recently used.

Reference :- 6, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 1, 2, 0
String

Assume 3 frames and frames are initially empty.

6	6	0	2	2	2	0	4	4	4	0	0	0	2
	0	0	0	0	0	0	0	0	3	3	3	3	3
		1	1	1	3	3	3	2	2	2	3	1	1
*	*	*	*		*		*	*	*	*		*	*
				Hit(0)		Hit(0)					Hit(3)		

Page Hit $\Rightarrow 3$

page Fault $\Rightarrow 12$

$$\text{Hit ratio} = 3/15 \times 100 = 20\% \quad \text{Fault ratio} = 12/15 \times 100 = 80\%$$

③ Optimal page Replacement

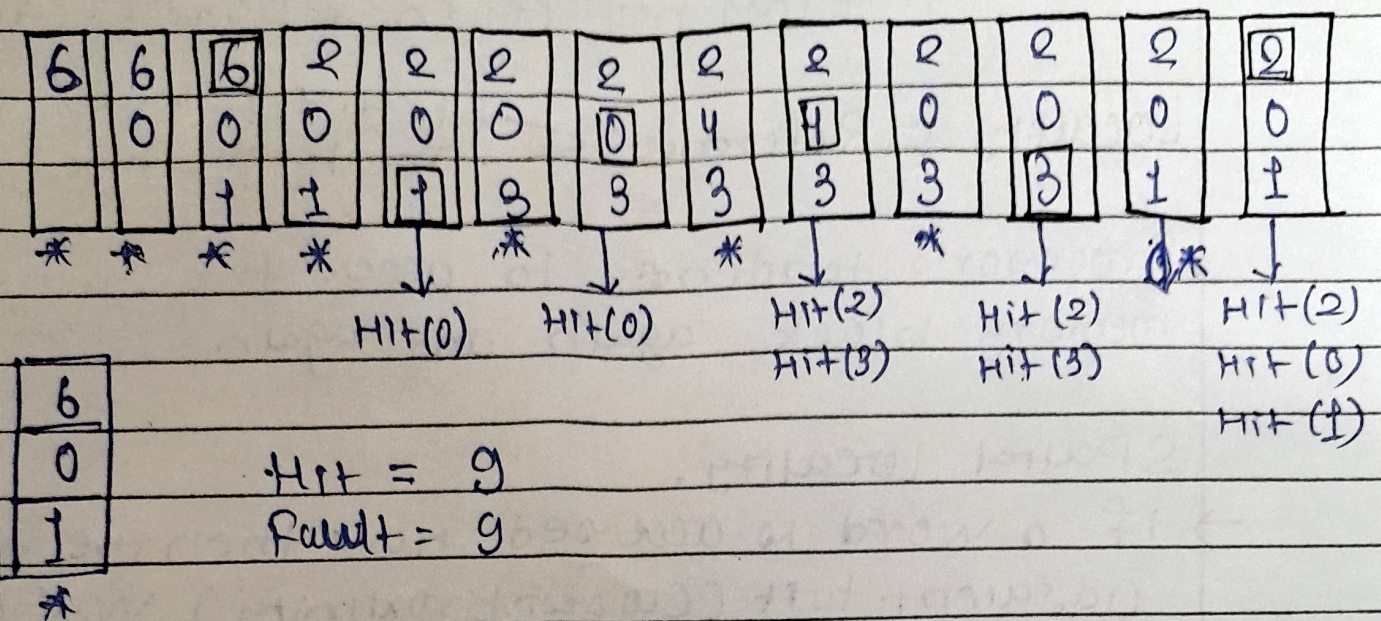
① Selects the pages for replacement that will be used very later in future

② Not possible in real life as this requires future knowledge

Same Question as above

Reference :- 6, 0, 1, 2, 0, 3, 0, 4, 3, 3, 0, 3, 2, 1, 2, 0, 1, 6
 String

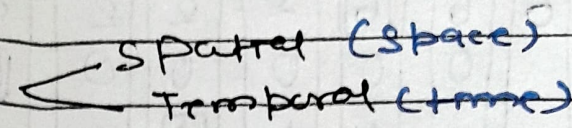
Assume 3 frames and frames are initially empty.



Cache Memory Organisation.

- Cache memory is high speed volatile computer memory
 - Stores frequently used computer programs
 - Crucial for improving optimal performance
 - Minimizes the time to access data and instructions
 - 3 levels of cache in a modern computer system
- ① L1 cache: Smallest, fastest, on CPU chip, separate instruction and data caches for independent handling
 - ② L2 cache: Slightly larger, slower than L1, often shared among different cores in a multi-core processor, provides additional storage for frequently accessed data.

- ⑧ L3 Cache → Larger, Slower than L2, Shared among all cores in a multi core processor, additional storage for frequently used data that may not fit in L1 and L2 caches.

Locality of Reference 

Processor's tendency to access the same memory block again and again.

Spatial Locality.

- If a word is accessed now then the word adjacent to it (Closest proximity) will be accessed next.
- Keeping words more in a block affects spatial locality.

Temporal Locality.

- If a word is referenced now then same word will be referenced again in future.
- LRU is used in temporal locality.